

3. If the URL is on the blacklist, the request is dropped.
4. If the URL is not on the blacklist, a new connection is initiated from IPCop to the destination Web server.
5. A reply from the Web server is terminated on IPCop.
6. IPCop forwards the reply to the corresponding Web browser.

Here, please observe that the traffic is decrypted and re-encrypted by IPCop. This results in a slight reduction in browsing speed, and technically, IPCop can read https traffic. But this is the way all https URL filters work.

Real life scenarios

Here are some real life scenarios where a properly configured firewall could have helped ...

Sasser worm: Sasser attacks the vulnerable Microsoft SMB port 445. Once affected, the system shuts down in one minute, displaying a count-down window. If you are really interested in experiencing Sasser, try this: Install Windows XP without the service pack, and connect your system to the Internet directly – in a short while, the system will get infected by Sasser. (Be very careful while you play around!) This is because, some Sassers are still in the wild and try affecting vulnerable systems even today!

Here, putting your system behind a network-based firewall will definitely help. Port 445 of the internal system will not be accessible from the Internet and the worm will be blocked by the firewall before entering the Internal network.

CodeRed worm: On July 13, 2001, a number of computer systems running the Microsoft IIS Web server on Windows NT or Windows 2000 were affected by a worm called CodeRed. It is believed that about 300,000 servers were affected on Day One! The worm was designed in a very peculiar manner. From the 1st of any

month to the 13th, it would try to spread to random IIS Web servers. Once a Web server was infected, it would in turn generate a lot of traffic towards the Internet searching for IIS servers randomly, ultimately infecting the vulnerable ones.

A properly configured firewall denying any traffic to be initiated from the Web server towards the Internet could have curtailed the spread of this worm. Also, a Web server should only reply to the incoming Web requests and should not be allowed to browse the Internet.

In this case, infection of vulnerable IIS Web servers (even those sitting behind a firewall) was inevitable, but a properly configured firewall could have definitely stopped the spread of CodeRed from the affected Web server.

ISP connectivity issues: In an earlier time, one ISP started its services from a central area of Pune with much fanfare. A horrifying fact was that all connected networks were not separated at the ISP end – in simple terms, double clicking on *Network Neighbourhood* made all the systems connected to this ISP visible and, of course, the data on them accessible! Here, deploying a firewall would have assured separation .

References

- [1] Google search for IPCOP LFY
- [2] IPCop official website: www.ipcop.org
- [3] A great IPCop forum: www.ipcops.com

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